

Image Processing Addon Bundle 3.2

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Class TIEProc_EX

TImageenproc ← TIEProc_EX
Declared in Unit NWSComps_Proc

What's new in v.3.2:

1. We have further improved the speed of the preview of the most time-consuming filters.
2. New TIEProc_EX_FiltersPanel that will build automatically the GUI to preview and apply the filters, including all sliders, checkboxes etc..you do not have to build anymore the user interface for the filters
3. We have solved some issues that could in some case give problems with the preview
4. We have adapted the component to work best with the latest version of ImageEn
5. Added a Color Filter, to apply a Color with different blend modes
6. Added a Bilateral Filter (denoising)
7. Added several other imageEn filter such as Median Filter (in Median and Sharpen Mode) and Sharpen

Description

TIEProc_EX is a visual component (descendant from TImageEnProc).

It has been designed with the aim of extending the native collection of imageEn filters, and providing real time preview for both its own filters and the native ImageEn filters. Furthermore, starting from v.3 of our bundle the component supports MultiThreaded Processing in order to maximize the use of modern computer cpus.

The ImageEn graphics library is needed to use all our components, and you can purchase it here: www.imageen.com

Features:

You can use the component just as a normal TImageEnProc, by setting its AttachedIEBitmap, or AttachedImageenview and call its methods exactly as in TImageenproc.

The advantage over a normal TImageenproc is that you can use multithreaded processing (25 ImageEn filters support multithread). You can use custom filters to encapsulate your own filter or imageEn filters that are not currently supported yet (see the demo on how to create a custom filter).

When multithreading is not supported, the single-threaded processing will be used automatically. You can enable / disable multithreading at the component level, or if you want to define your own rule at the filter level you can use the provided OnFilter_CustomMultithreadInfo event.

You can create filters / collection of filters, either from our predefined supported types that includes 28 native imageEn filters, or from your own custom filters (see the demo project). You can apply a single filter or a collection of filters, in the same way you would do using a TImageenproc, thanks to the Process_Apply method.

Using the advanced approach with filters you can preview any filter / collection of filters in real time. Filters can be easily previewed in real time on any TImageEnView component. All you have to do, is to call the Preview_Register method to register the component together with the display (TImageEnView) and with the Filter or Collection of filters you want to preview.

The new Filter Panel component builds the GUI for the filters automatically (no more adding trackbars, checkboxes etc.. to your project, all is done by this component)

During the preview the user will be able to change the filters parameters and the preview will be refreshed. Then you can apply the preview, just calling a method of the component (Preview_Apply). The demo project shows how to do that with minimal coding.

The preview will work on the currently selected layer in TImageEnView. If you change the current layer, the preview also switches automatically on the new layer. The preview will take also into account any selection that is present in the ImageEnView component. And it will be refreshed automatically each time there is a change to the display (zoom, pan, etc..). This is done completely automatically, you do not have to write any code, except registering TIEProc_Ex with your Timageenview by a single method call.

All other not-included ImageEn filters can be previewed or you can add multithreading support to them but you have to encapsulate them into custom filters. Several examples of custom filters are offered in the demo project.

TIEProc_EX includes also our own filters that are not present in ImageEn such as: Smart Flash, Reduce Highlights, Fill-Back Light, Smart Contrast. These can be both previewed and applied as all the other filters.

Using custom filters, you can now use TIEProc_EX together with our TRGBCurves component. An example is provided in the demo project.

Technical Reference

UNITS:

NWSComps_Proc.pas

Contains the main class TIEProc_Ex that is derived from TImageEnProc

NWSComps_Proc_Engine.pas

Contains the classes and functions of the image processing engine.

NWSComps_Proc_Filter_Types.pas

Contains the general classes for filters and filters collections.

NWSComps_Proc_Filter_Lib

Contains the specific classes for the kind of filters actually supported.

NWSComps_Proc_Filter_Lib_const.pas

Contains all the constants needed to define the parameters of the filters

NWSComps_MultiThreadProc.pas

It is the specialized module for multithreaded processing.

TIEProc_EX

Properties

property MultiThreadEnabled: boolean

Defines the enabling / disabling of Multithreading at component's level. When this is disabled, enabling multithreading at filter level, will not have effect.

property MultiThreadNrThreads: cardinal

Defines the number of Threads to be used in Multithreading at component's level. This value will be used for all filters, unless otherwise defined in the OnFilter_CustomMultiThreadInfo event handler.

property MultiThreadDefOverlapMethod: TIEMultiProc_EX_OverlapMethod

Defines the Overlap Method for multithreading at component's level. This value will be used for all filters, unless otherwise defined in the OnFilter_CustomMultiThreadInfo event handler.

property UseFastPreview: boolean

Defines the use of fast preview algorithm. It is best leave this on

property PreviewMode: TIEUtils_IEPreview_Mode

Can have 3 values:

1. pm_Auto: determines the preview mode automatically, choosing from the two other options listed below. It is best to choose this option in any case
2. pm_Flat: to use only when the picture has no layers and no selections
3. pm_FullUpdate: this is the most common type of preview, used when picture can have layers and selections

Methods

procedure Preview_Register(theIEView: TImageenview; theFilter: TIEProc_EX_Filter);overload;

Registers the preview of a single filter together with specified Imageenview

procedure Preview_Register(theIEView: TImageenview; theFilterList: TIEProc_EX_Filter_Collection);overload;

Registers the preview of a collection of filters together with specified Imageenview

procedure Preview_Unregister;

Call this method once you are done with the preview, to disable any update of the preview on the display.

procedure Preview_Apply;

This applies the filter or collection of filters that is currently being previewed to the current layer of the TImageEnview display. It should be used while still in preview mode.

procedure Preview_Lock; procedure Preview_Unlock;

These two methods lock and unlock the preview (while in preview mode), in order to change the content of the TImageEnvview, for example loading a new picture or layer).

procedure Preview_Toggle(const bPreviewIsOn: boolean);

This method allows to switch on/off the preview while in preview mode in order to see the difference between the original picture and the picture with the filter applied.

**procedure Process_Apply(theFilter: TIEProc_EX_Filter; const iFilter, iTot: integer)
procedure Process_Apply(theFilterList: TIEProc_EX_Filter_Collection);**

Call these methods to apply any Filter or filter collection to the attached TIEBitmap / TBitmap / TImageEnvview but without previewing them.

procedure Process_ApplyCurrent;

Call this method to apply the currently registered Filter or collection of filters to the attached TIEBitmap / TBitmap / TImageEnvview but without any preview.

Events

OnFilter_Progress: TIEProgressEvent

Use this with a progress bar to see the advancement of the filter processing

OnFilter_FinishWork: TnotifyEvent

This event occurs when the filter process is finished

OnFilter_CustomMultiThreadInfo: TIEProc_Ex_CustomMultiThreadInfoEvent

TIEProc_Ex_CustomMultiThreadInfoEvent is defined as follows:

**procedure(filter: TIEProc_EX_Filter;
var MtEnabled: TIEProc_EX_Filter_MultiThreadEnabled; var MtNrThreads: cardinal;
var theOverlap: cardinal; var bAutoOverlap: boolean;
var OverlapMethod: TIEMultiProc_EX_OverlapMethod) of object;**

→ **Filter**: The filter for which you want to specify the custom Multithreading info.

→ **MtEnabled** specify if your filter will accept Multithreading. It can have one of the following values:

- mtAlwaysEnabled
- mtEnabledExceptPreview (some default filters do require MT to be disabled when previewing)
- mtDisabled

→ **MtNrThreads**: The number of threads you want to use for the current filter

→ **theOverlap**: The overlap in pixels you want to use for the current filter. It is 0 for all filters that compute the output from the input in a 1 to 1 pixel relation. In case your filter computes the output based on the values of the surrounding pixels, the bigger the radius in which the filter looks for, the bigger must be this value. The overlap value is already optimized for all the filters supported. Do not change this value if unsure

→ **bAutoOverlap**: specify if the Multithreading engine is allowed to change the overlap for its own

optimizations. The AutoOverlap value is already optimized for all the filters supported. Do not change this value if unsure.

→ **OverlapMethod:** Specify the overlap method to be used for the current filter. There are two possible values:

- omAddExtraPixels: achieves MT by adding extra pixels to process
- omAddExtraThreads: achieves MT by adding extra threads to process

List of Own Filters:

TIEProc_EX_Filter_SmartFlash

Multithreaded: YES

This filter acts on the dark (underexposed) parts of the picture, by smoothly illuminating them, without affecting the other parts of the picture that are already correctly exposed. It acts like an intelligent digital flash light.

Parameters:

Amount (integer: 0-255)

Radius (Integer)

TIEProc_EX_Filter_ReduceHighlights

Multithreaded: YES

This filter acts on the bright (overexposed) parts of the picture, by smoothly lowering them to the desired level, without affecting the other parts of the picture that are already correctly exposed. It can recover tones and details lost in the overexposed parts of the picture.

Parameters:

Amount (integer: 0-255)

Radius (Integer)

TIEProc_EX_Filter_FillBackLight

Multithreaded: YES

This filter improves the shadows and the high-lights, with the effect of recalibrating the tone-range of the picture.

Parameters:

FillLightAmount (integer: 0-255)

BackLightAmount (integer: 0-255)

Radius (Integer)

TIEProc_EX_Filter_SmartContrast

Multithreaded: YES

This filter improves the contrast and details of the picture, especially in the flat areas of the picture. It gives a sharper appearance, with clearer details to the picture.

Parameters:

Amount (integer: 0-255)
Radius (Integer)

TIEProc_EX_Filter_RGBBalance

Multithreaded: YES

This filter provides a way to change the Color of an image using weights for different parts of the tonal range.

Parameters:

Uniform (boolean)
Amount Shadows Red (integer: -255 / +255)
Amount Shadows Green (integer: -255 / +255)
Amount Shadows Blue (integer: -255 / +255)
Amount Midtones Red (integer: -255 / +255)
Amount Midtones Green (integer: -255 / +255)
Amount Midtones Blue (integer: -255 / +255)
Amount Highlights Red (integer: -255 / +255)
Amount Highlights Green (integer: -255 / +255)
Amount Highlights Blue (integer: -255 / +255)
Amount All Tones Red (integer: -255 / +255)
Amount All Tones Green (integer: -255 / +255)
Amount All Tones Blue (integer: -255 / +255)

TIEProc_EX_Filter_AutoColor

Multithreaded: Disabled

This filter can be used to automatically balance the color in a picture and removing light color casts. The effect of this filter is subtle and it can be regulated by the strength parameter.

Parameters:

Strength (integer: 1-10)

TIEProc_EX_Filter_ColorFilter

Multithreaded: YES

This filter applies a color to the image using different blend modes.

Parameters:

Amount (0-255)
Color:TColor
BlendMode: TIEProc_EX_Filter_Blendmode = (blmnormal = 0,
blmoverlay = 1,
blmhardlight = 2,
blmmultiply = 3,
blmscreen = 4);

TIEProc_EX_Filter_BilateralFilter

Multithreaded: YES

This filter applies the bilateral algorithm that can be used to denoise images. It can be very slow on big images.

Parameters:

Sigma: typically between 10 and 40, this is the amount of denoising you apply

Sigma Distance: the distance in the gaussian exponential algorithm. Typically between 10 and 20

Radius: the increase of this parameters tends to make the filter slow. For small images use 1 or 2 for bigger images 3 or 4.

TIEProc_EX_Filter_Smooth

Multithreaded: YES

This filter smooths the image preserving the edges

Parameters:

Amount: the amount of the filter to be applied between 1 and 80

TIEProc_EX_LightFX_Ellipse

Multithreaded: YES

Creates an elliptical light effect.

Parameters:

Color: The color of the Light (Tcolor)

Light Amount: the amount of the light between 0 and 100

Ellipse Minor Axis: between 1 and 10000 (you can change the max value at runtime)

Ellipse Major Axis: between 1 and 10000 (you can change the max value at runtime)

Angle: The angle: between 0 and 360

CenterX: the X coordinate of the center 1..10000 (you can change the max value at runtime)

CenterY: the Y coordinate of the center 1..10000 (you can change the max value at runtime)

TIEProc_EX_LightFx_Beam

Multithreaded: YES

Creates a beam of light effect.

Parameters:

Color: The color of the Light (Tcolor)

Light Amount: the amount of the light between 0 and 100

Beam Size: 1.5000 (you can change the max value at runtime)

Angle: The angle: between 0 and 360

CenterX: the X coordinate of the center 1..10000 (you can change the max value at runtime)

CenterY: the Y coordinate of the center 1..10000 (you can change the max value at runtime)

TIEProc_EX_LightFx_DoubleBeam

Multithreaded: YES

Creates a double beam of light effect.

Parameters:

Color: The color of the Light (Tcolor)
Light Amount: the amount of the light between 0 and 100
Beam Size: 1.5000 (you can change the max value at runtime)
Angle: The angle: between 0 and 360
Opening: between 0 and 100
CenterX: the X coordinate of the center 1..10000 (you can change the max value at runtime)
CenterY: the Y coordinate of the center 1..10000 (you can change the max value at runtime)

List of natively supported ImageEn Filters:

TIEProc_EX_Filter_IE_UnsharpMask

Multithreaded: Yes

TIEProc_EX_Filter_IE_Blur

Multithreaded: Yes

TIEProc_EX_Filter_IE_HSLVar

Multithreaded: Yes

TIEProc_EX_Filter_IE_HSVVar

Multithreaded: Yes

TIEProc_EX_Filter_IE_Negative

Multithreaded: Yes

TIEProc_EX_Filter_IE_IntensityRGBAll

Multithreaded: Yes

TIEProc_EX_Filter_IE_Intensity

Multithreaded: Yes

TIEProc_EX_Filter_IE_Contrast

Multithreaded: Yes

TIEProc_EX_Filter_IE_Contrast2

Multithreaded: Yes

TIEProc_EX_Filter_IE_Contrast3

Multithreaded: Yes

TIEProc_EX_Filter_IE_GammaCorrect

Multithreaded: Yes

TIEProc_EX_Filter_IE_AutoSharp

Multithreaded: Yes

TIEProc_EX_Filter_IE_Colorize

Multithreaded: Yes

TIEProc_EX_Filter_IE_AdjustBrightnessContrastSaturation

Multithreaded: Yes

TIEProc_EX_Filter_IE_AdjustTint

Multithreaded: Yes

TIEProc_EX_Filter_IE_AdjustSaturation

Multithreaded: Yes

TIEProc_EX_Filter_IE_AdjustTemperature

Multithreaded: Yes

TIEProc_EX_Filter_IE_AdjustLumSatHistogram

Multithreaded: Yes

TIEProc_EX_Filter_IE_AutoImageEnhance1

Multithreaded: Disabled

TIEProc_EX_Filter_IE_AutoImageEnhance2

Multithreaded: Disabled

TIEProc_EX_Filter_IE_AutoImageEnhance3

Multithreaded: YES

TIEProc_EX_Filter_IE_AdjustLumSatHistogram

Multithreaded: YES

TIEProc_EX_Filter_IE_WhiteBalance_coef

Multithreaded: YES

TIEProc_EX_Filter_IE_WhiteBalance_Grayworld

Multithreaded: Disabled

TIEProc_EX_Filter_IE_WhiteBalance_WhiteAt

Multithreaded: YES

TIEProc_EX_Filter_IE_WhiteBalance_AutoWhite

Multithreaded: Disabled

TIEProc_EX_Filter_IE_Median_Sharpen

Multithreaded: Yes

This is the Imageen filter: Median, but used in Sharpen mode (there are 3 modes in the original filter)

TIEProc_EX_Filter_IE_Median

Multithreaded: Yes

This is the Imageen filter: Median, used in Median mode (there are 3 modes in the original filter)

TIEProc_EX_Filter_IE_Sharpen

Multithreaded: Yes

Class TIEProc_EX_FiltersPanel

TScrollbar ← **TIEProc_EX_FiltersPanel**

Declared in Unit NWSComps_FiltersPanel

This component lets you build the user interface for the filters in no time. For example:

```
fFilterList := TIEProc_EX_Filter_Collection.Create; //Create the collection of filters
```

//now we add three filters to the list

```
fFilterList.Add(TIEProc_EX_Filter_IE_AdjustTemperature.Create);
fFilterList.Add(TIEProc_EX_Filter_IE_HSVVar.Create);
fFilterList.Add(TIEProc_EX_Filter_IE_UnsharpMask.Create);
```

//now we assign the filters to the FiltersPanel:

```
FiltersPanel1.FilterList := fFilterList; //Creates automatically the filters GUI
```

Every time you assign the FilterList property the GUI is rebuilt, so once done, do it again only if you want to assign another filter list or you have changed the number or the order of the filters in the same filter list.

//Now for example we reset the values of the filters

```
for I := 0 to fFilterList.Count-1 do
begin
  fFilterList.Filter[i].Active := false; //set active to false
  for j := 0 to filter.Params.Count-1 do
  begin
    param := filter.Params.Params[j];
    if param.Has_Default then
      filter.SetParamValue(param.Name, param.DefValue); //set default value
    end;
  end;
end;
```

//And update the filters Panel just calling a method

```
FiltersPanel1.UpdateFromFilters;
```

Now if you want to start to preview the filters on a Timageenview you only have to register the TIEProc_Ex component.

```
IEProcEX1.Preview_Register(ImageEnView1, fFilterList);
```

Then using the UI provided by the FiltersPanel you will be able to preview the effect of the filters on the registered Imageenview.

Events

The FiltersPanel has also some useful events:

OnAddFilterParam: TIEProc_EX_FiltersPanelAddParamEvent;

You can use this event to exclude some parameters from the user interface. So that they cannot be manipulated by the user.

For example:

```
FiltersPanel1.OnAddFilterParam := HandleAddFilterParam;
```

```
procedure TForm1.HandleAddFilterParam(Filter:TIEProc_EX_Filter; Param:TIEProc_EX_Filter_Param;
var bCanAdd:boolean);
```

```
begin
  //by default bCanAdd is true
  if Filter.ID = IEPROC_EX_IE_AUTOIMAGEENHANCE1_ID then
  begin
    bCanAdd := false; //no parameters for autoenhance (use default)
  end
  else if Filter.ID = IEPROC_EX_IE_AUTOIMAGEENHANCE2_ID then
  begin
    bCanAdd := false; //no parameters for autoenhance (use default)
```

```

end
else if Filter.ID = IEPROC_EX_IE_AUTOIMAGEENHANCE3_ID then
begin
    bCanAdd := false; //no parameters for autoenhance (use default)
end
else if Filter.ID = IEPROC_EX_IE_GAMMACORRECT_ID then
begin
    //we want only the Gamma amount
    bCanAdd := Param.Name = IEPROC_EX_IE_GAMMACORRECT_GAMMA;
end;
end;
end;

```

OnCustomInsert: TIEProc_EX_FiltersPanelCustomInsertEvent;

This event is useful if you want to add a custom control for a custom filter, such as our TRGBCurves control. For example:

```
FiltersPanel1.OnCustomInsert := HandleFiltersPanelCustomInsert;
```

```

procedure TForm1.HandleFiltersPanelCustomInsert(Filter:TIEProc_EX_Filter; var ControlToInsert:
TControl);
begin
    if Filter.ID = CUSTOMFILTER_RGBCURVES then
        ControlToInsert := Panel_Curves;
end;

```